

- Document Control
- System Name: Application
- Version: 1.0
- Author: Automated SDLC Generator
- Date: 2024-06-12
- Status: Draft
- Executive Summary
 - The Application facilitates an automated ETL pipeline involving source CSV data ingestion, transformation, cleaning, aggregation, and SCD Type 2 history management, implemented using Delta Live Tables.
- Business Objectives
 - Automate data ingestion from CSV files to Delta tables
 - Ensure data quality through null and duplicate removal
 - Aggregate customer spend and maintain accurate, auditable transaction history
 - Enable change data capture via SCD Type 2 implementation
- In-Scope
 - Source CSV data reading and delta table loading
 - Schema enforcement for customer and order tables
 - Data transformation and cleaning logic
 - Creation of ordersummary and customeraggregatespend tables
 - Aggregation logic on ordersummary data
 - SCD Type 2 implementation for ordersummary
 - Process deployment using Delta Live Tables
- Out-of-Scope
 - Any manual data manipulation
 - External system integrations beyond defined delta tables
 - Real-time or streaming ingestion
 - Data visualization or reporting dashboards
- Stakeholders
 - Data Engineering Team
 - Business Analysts
 - Compliance Team
 - Data Platform Operations

- Business Requirements

BR-ETL-001 — Load Source Customer Data

- Description: Read customer CSV data and load to delta table /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata
- Business Value: Automated, structured customer data ingestion
- Priority: High

BR-ETL-002 — Load Source Order Data

- Description: Read order CSV data and load to delta table /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata
- Business Value: Automated, structured order data ingestion
- Priority: High

BR-ETL-003 — Customer Table Schema Enforcement

- Description: Enforce schema with fields CustId, Name, EmailId, Region for customer table
- Business Value: Consistency and data quality assurance
- Priority: High

BR-ETL-004 — Order Table Schema Enforcement

- Description: Enforce schema with fields OrderId, ItemName, PricePerUnit, Qty, Date, CustId for order table
- Business Value: Consistency and data quality assurance
- Priority: High

BR-ETL-005 — Calculate TotalAmount Field

- Description: Add column TotalAmount to order table ($\text{PricePerUnit} * \text{Qty}$)
- Business Value: Simplifies downstream aggregation and analytics
- Priority: Medium

BR-ETL-006 — Data Quality: Null and Duplicate Removal

- Description: Remove nulls and duplicates from customer and order tables
- Business Value: Improves analytical accuracy and reliability
- Priority: High

BR-ETL-007 — Create Ordersummary Table

- Description: Create ordersummary table in catalog=gen_ai_poc_databrickscoe, schema=sdlc_wizard if not exists

- Business Value: Centralized storage for enriched order data
- Priority: High

BR-ETL-008 — SCD Type 2 on Ordersummary

- Description: Apply SCD Type 2 methodology to ordersummary based on updates in customer table
- Business Value: Ensures historical accuracy and traceability of data changes
- Priority: High

BR-ETL-009 — Create CustomerAggregateSpend Table

- Description: Create customeraggregatespend table with columns Name, TotalAmount, Date if not exists
- Business Value: Enables spend analysis and reporting
- Priority: Medium

BR-ETL-010 — Aggregate Customer Spend

- Description: Aggregate TotalAmount grouped by Name and Date from ordersummary and load into customeraggregatespend
- Business Value: Facilitates spend tracking for business insights
- Priority: Medium

BR-ETL-011 — Delta Live Tables Implementation

- Description: Implement the entire processing steps using Delta Live Tables
- Business Value: Simplifies workflow orchestration and monitoring
- Priority: High
- Assumptions & Constraints
- Source data is available and accessible via defined CSV file paths
- Only the specified schemas will be enforced
- Delta Lake infrastructure is operational
- Sufficient permissions and quota for processing jobs
- Success Metrics
- Successful, automated runs from CSV to target tables
- Zero null/duplicate records post-processing
- Accurate aggregation results in customeraggregatespend
- SCD Type 2 audit trails present
- Risks & Mitigations

- Source data format changes — Mitigation: Schema enforcement and validation
- Data quality issues from source — Mitigation: Robust null and duplicate handling logic
- Infrastructure/permission errors — Mitigation: Pre-execution checks and monitoring
- Job runtime overruns — Mitigation: Performance monitoring and auto-scaling

- Approval
- Business Owner: Data Platform Lead
- Data Engineering Lead: Senior Data Engineer
- Compliance Review: Data Governance Officer
- Date: 2024-06-12

- Appendix

Appendix A: Source Definitions

Source Name	Type	Location	Format	Notes
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customerdata	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customer.csv	CSV	Customer master feed
orderdata	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/order.csv	CSV	Order transaction feed

Appendix B: Target Definitions

Target Name	Layer	Table	Columns	Notes
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customerdata	Bronze	customerdata	CustId, Name, EmailId, Region	Post-ingestion customer staging table
orderdata	Bronze	orderdata	OrderId, ItemName, PricePerUnit, Qty, Date, CustId, TotalAmount	Post-ingestion order staging table
ordersummary	Silver	ordersummary	OrderId, ItemName, TotalAmount, Qty, Date, CustId, ValidFrom, ValidTo, IsCurrent	Enriched order data with SCD2 tracking
customeraggregatespend	Silver	customeraggregatespend	Name, TotalAmount, Date	Aggregated customer spend

Appendix C: Transformations

Step	Transformation	Input	Output	Logic
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- 1 | Add TotalAmount | orderdata | orderdata | PricePerUnit * Qty
- 2 | Remove Nulls | customerdata, orderdata | customerdata, orderdata | Drop records with nulls in critical fields
- 3 | Remove Duplicates | customerdata, orderdata | customerdata, orderdata | Drop duplicate records
- 4 | SCD2 Updates | ordersummary | ordersummary | Apply SCD Type 2 for historical tracking
- 5 | Aggregate Spend | ordersummary | customeraggregatespend | Group by Name, Date, sum(TotalAmount)

Appendix D: Business Rules

Rule ID | Area | Description | Source

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BR-ETL-003 | Data Quality | Enforce schema for customer table | BRD

BR-ETL-004 | Data Quality | Enforce schema for order table | BRD

BR-ETL-006 | Data Quality | Remove nulls and duplicates post-ingestion | BRD

BR-ETL-008 | SCD2 | Apply SCD Type 2 changes to ordersummary | BRD

BR-ETL-010 | Aggregation | Aggregate spend per customer and date | BRD

Appendix E: Process Flow

Step | Description | Dependency

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1 | Ingest customer CSV file | None

2 | Ingest order CSV file | None

3 | Enforce schema, clean data | Steps 1, 2

4 | Add TotalAmount to order table | Step 3

5 | Create ordersummary table with SCD Type 2 | Step 4

6 | Aggregate spend to customeraggregatespend | Step 5